

Lab #4 - snort

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Summary: This lab was about configuring an intrusion detection system within a Linux environment in Labtainer. While completing this lab we all learned how Snort monitors network traffic, and how it creates custom rules for detecting specific types of traffic, and observes how encryption such as HTTPS affects the IDS's ability to analyze traffic. We also learned how to configure Snort for different scenarios. We got this done by simulating network scans and analyzing the behavior of Snort.

Background: In this lab we use an IDS which is a software application designed to monitor network or system activities for malicious activity or policy violations. Then we move on to the use of snort which is an open source IDS that can perform real time traffic analysis and packet logging. By doing this it is always on alert for attacks; it does this by inspecting network traffic and checking if anything is against its detection rules. The experiment described in this lab focuses on monitoring network traffic and detecting potential attacks using Snort IDS.

Tools and System Specifications: The tools and system specifications used for this lab are, labtainer, snort which is the prevention and detection system. The nmap which is used as a network scanning tool, firefox and linux.

What have you learned from this lab and how will it help you to protect the system from vulnerability and attacks?

In this lab we learned how to use the snort system to provide intrusion detection within a Linux environment. We were able to configure simple snort rules and also gained experience by experimenting with a network intrusion detection system. By implementing and testing the rules,

such as detecting “confidential” data, it helped us better understand how snort identifies suspicious patterns. We also learned that there are limitations when dealing with encrypted HTTPS traffic. Furthermore we learned how to mirror traffic using iptables. As this was our first experience working with snort, the steps we learned in this lab will help us better protect our systems from vulnerabilities.

Results, Discussions, and Conclusion:

1. Lab Environment Setup

1. Launch Labtainer and Start the Snort Lab:

Open a terminal in your labtainer-student directory.

```
labtainer snort
```

This command initializes the lab setup, creating virtual terminals for each component.

2. Review Network Configuration:

Review the network topology provided in the lab manual. Key components include:

Gateway: Handles NAT and traffic mirroring to the Snort IDS.

Snort Component: Monitors network traffic.

Web Server: Hosts a website configured with SSL.

Remote Workstations: External and internal workstations, each with access to network monitoring and testing tools.

2. Start and Stop Snort

1. Starting Snort in Network Intrusion Detection Mode:

In the Snort component terminal, start Snort by executing:

```
./start_snort.sh
```

Snort will begin running and display alerts directly in the console.

```
tom@snort: ~  
mary@ws2: ~  
tom@snort:~$ ./start_snort.sh
```

2. Stopping Snort:

When you need to stop Snort (e.g., to modify rules), press CTRL+C.

3. Understanding Pre-Configured Snort Rules

1. Run a Network Scan and Observe Alerts:

From the remote workstation, perform an nmap scan to trigger alerts:

```
sudo nmap www.example.com
```

Watch for alerts on the Snort console. Alerts generated by the preconfigured rules can be examined in the `/etc/snort/rules/` directory.

```
hank@remote_ws: ~  
hank@remote_ws:~$ sudo nmap www.example.com  
  
Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-02 21:32 UTC  
Nmap scan report for www.example.com (203.0.113.10)  
Host is up (0.00024s latency).  
Not shown: 996 closed ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 5.00 seconds  
hank@remote_ws:~$
```

```
tom@snort: ~  
mary@ws2: ~  
tom@snort:~$ ./start_snort.sh  
04/02-21:32:44.961675  [**] [1:469:3] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Priority: 2] {ICMP} 203.0.113.20 -> 203.0.113.10  
04/02-21:32:44.961675  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.20 -> 203.0.113.10  
04/02-21:32:44.963222  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.20 -> 203.0.113.10  
04/02-21:32:47.874279  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 203.0.113.20:64376 -> 203.0.113.10:705  
04/02-21:32:47.940581  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 203.0.113.20:64376 -> 203.0.113.10:161
```

4. Write and Test a Basic Custom Snort Rule

1. Create a Simple Rule to Detect All TCP Traffic:

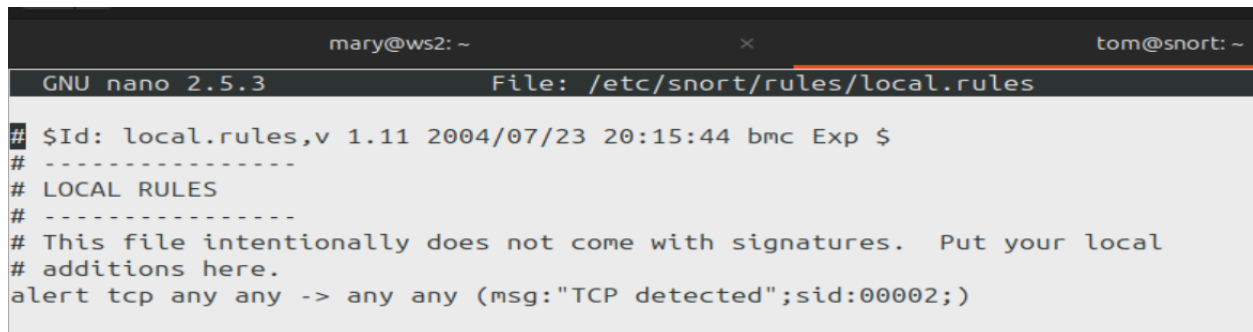
Stop Snort by pressing CTRL+C.

Open the custom rule file:

```
sudo nano /etc/snort/rules/local.rules
```

Add the following rule to alert on every TCP packet:

```
alert tcp any any -> any any (msg:"TCP detected"; sid:00002;)
```



```
mary@ws2: ~ x tom@snort: ~
GNU nano 2.5.3 File: /etc/snort/rules/local.rules
## $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert tcp any any -> any any (msg:"TCP detected";sid:00002;)
```

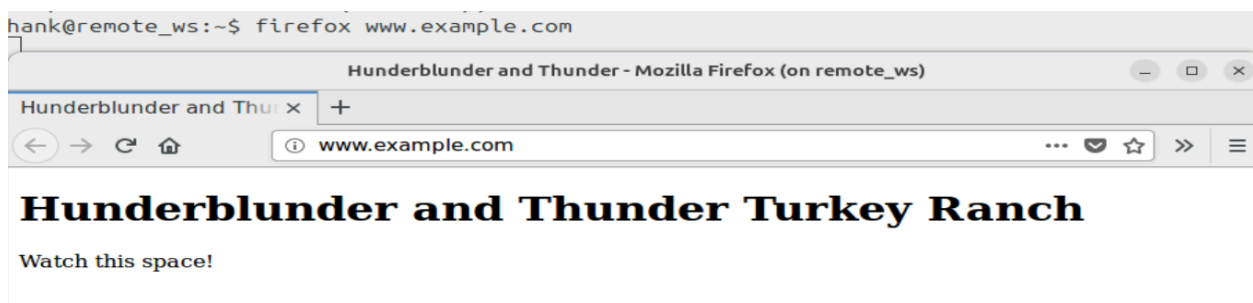
Save the file and restart Snort by running `./start_snort.sh`.



```
tom@snort:~$ ./start_snort.sh
```

2. Test the Rule:

Use Firefox on the remote workstation to visit www.example.com.



Observe the Snort console for numerous alerts. Once confirmed, stop Snort and delete this rule to avoid excessive alerts.

```

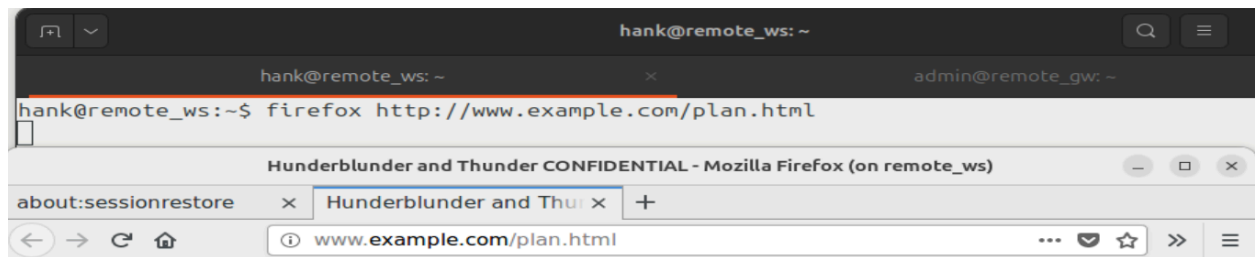
tom@snort:~$ ./start_snort.sh
04/02-21:42:54.367368  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:54.367678  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
04/02-21:42:54.368613  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:56.191878  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:56.192107  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
04/02-21:42:56.214354  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
04/02-21:42:56.214753  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:57.879102  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:57.882809  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686

```

5. Create a Custom Rule to Detect "CONFIDENTIAL" Traffic

1. View the Confidential Page:

Use Firefox to open <http://www.example.com/plan.html> on the remote workstation to see a confidential document.



Hunderblunder and Thunder Turkey Ranch

Startup Business Plan: CONFIDENTIAL

First we get some turkey eggs and figure out how to hatch them. Then we get some turkey feed. Then we find an ax. OK, I did my bit, you guys run with it.

2. Add a Rule to Detect Confidential Content:

Open the custom rule file and add a rule to trigger an alert when

"CONFIDENTIAL" is detected:

```

alert tcp any any -> any any (msg:"CONFIDENTIAL content detected";
content:"CONFIDENTIAL"; sid:00003;)

```

```

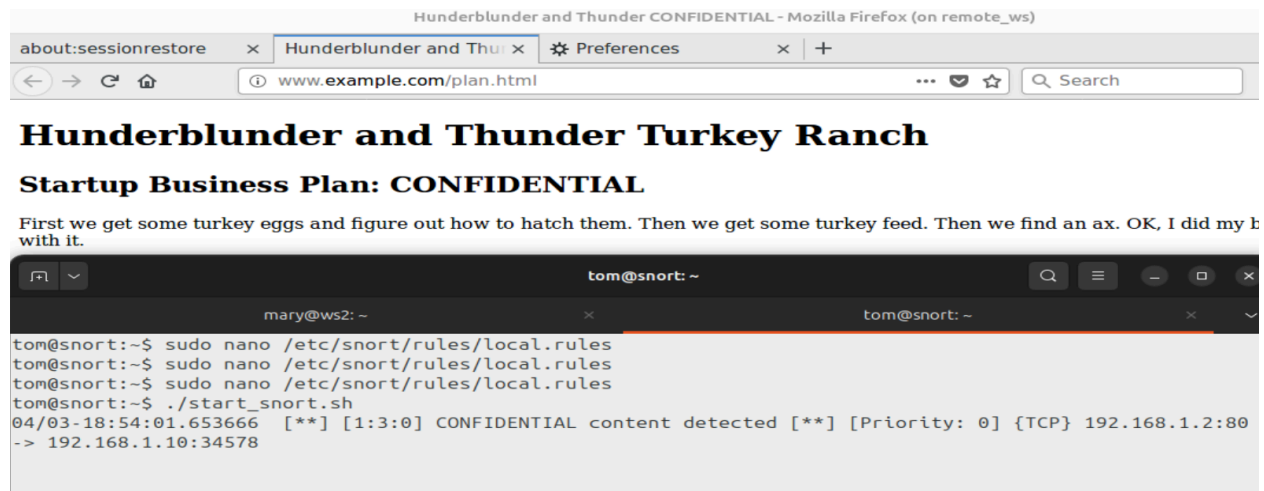
## $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> any any (msg:"CONFIDENTIAL content detected"; content:"CONFIDENTIAL"; sid:00003;)

```

Restart Snort and clear the Firefox browsing history (Menu > Preferences >

Security & Privacy).

Refresh plan.html in Firefox to trigger the alert.

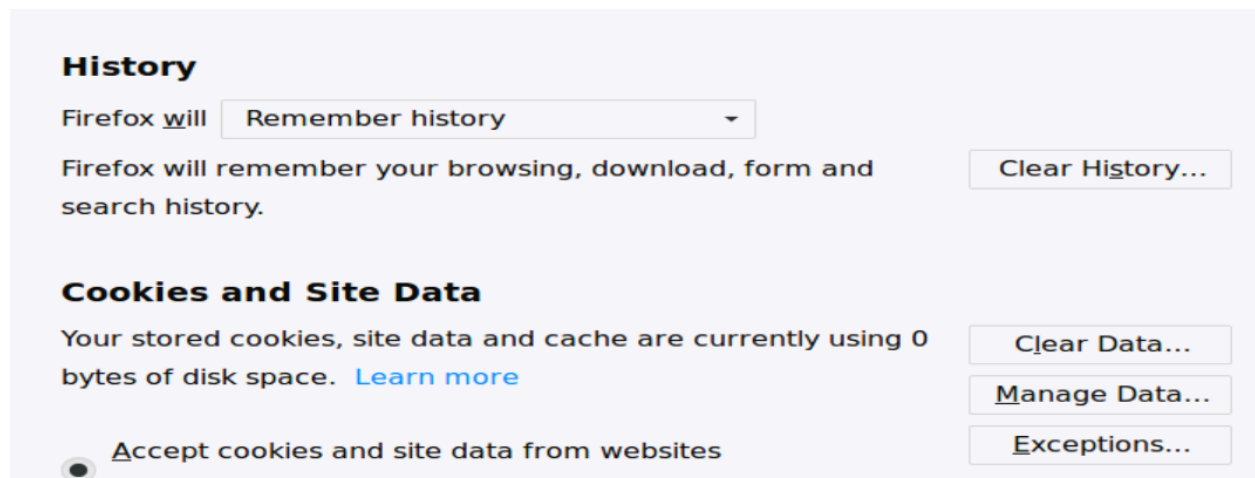


6. Explore Effects of Encryption on Intrusion Detection

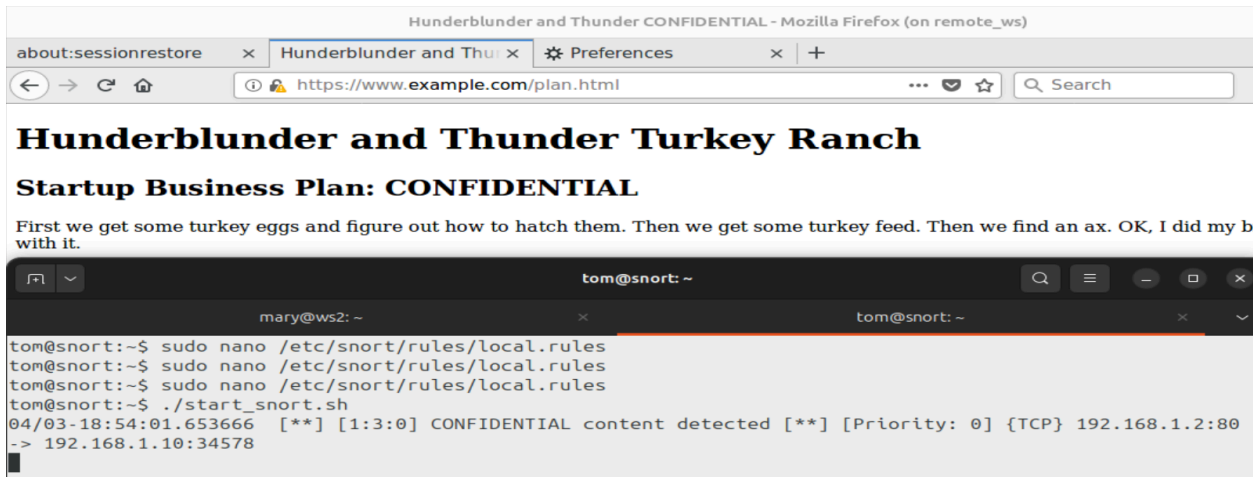
1. Access the Confidential Page via HTTPS:

Clear the browsing history again, then navigate to

<https://www.example.com/plan.html>.



Observe the Snort console. You should not see a new alert because HTTPS encrypts the content, making it unreadable by Snort.



7. Monitor Internal Traffic

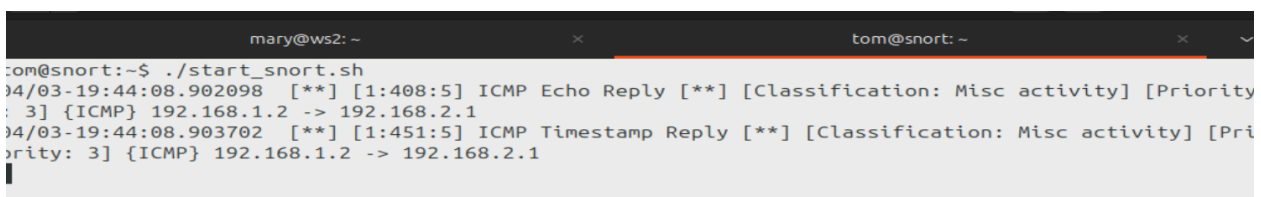
1. Run nmap on an Internal Workstation (ws2):

Go to the ws2 terminal and execute:

`sudo nmap www.example.com`



No alert should appear on the Snort console because internal traffic is not yet mirrored.



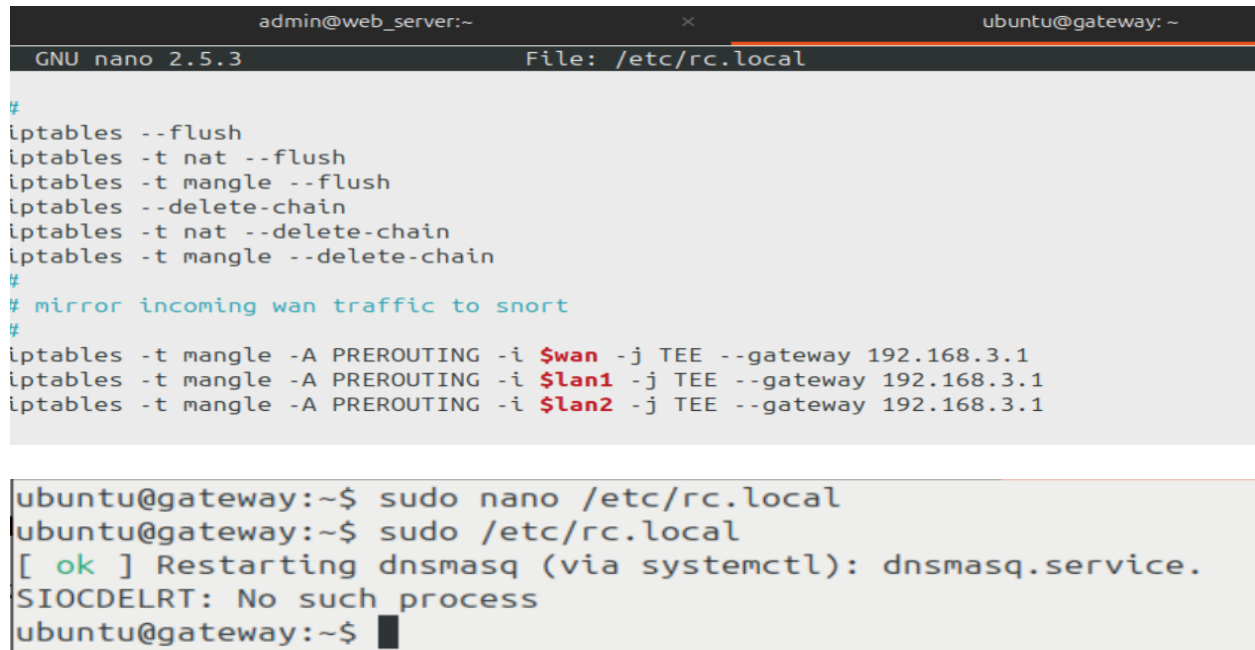
2. Mirror Internal Traffic to Snort:

On the gateway component, edit the /etc/rc.local file to add the following line:

```
iptables -t mangle -A PREROUTING -i $lan2 -j TEE --gateway  
192.168.3.1
```

Apply the new settings:

```
sudo /etc/rc.local
```



```
admin@web_server:~  
ubuntu@gateway: ~  
GNU nano 2.5.3 File: /etc/rc.local  
#  
iptables --flush  
iptables -t nat --flush  
iptables -t mangle --flush  
iptables --delete-chain  
iptables -t nat --delete-chain  
iptables -t mangle --delete-chain  
#  
# mirror incoming wan traffic to snort  
#  
iptables -t mangle -A PREROUTING -i $wan -j TEE --gateway 192.168.3.1  
iptables -t mangle -A PREROUTING -i $lan1 -j TEE --gateway 192.168.3.1  
iptables -t mangle -A PREROUTING -i $lan2 -j TEE --gateway 192.168.3.1  
  
ubuntu@gateway:~$ sudo nano /etc/rc.local  
ubuntu@gateway:~$ sudo /etc/rc.local  
[ ok ] Restarting dnsmasq (via systemctl): dnsmasq.service.  
SIOCDELRT: No such process  
ubuntu@gateway:~$
```

Restart Snort, then rerun the nmap scan from ws2 to see alerts generated for internal traffic.



```
mary@ws2:~$ sudo nmap www.example.com  
  
Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-03 20:02 UTC  
Nmap scan report for www.example.com (192.168.1.2)  
Host is up (0.00036s latency).  
Not shown: 996 closed ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
443/tcp   open  https  
3306/tcp  open  mysql  
  
Nmap done: 1 IP address (1 host up) scanned in 7.69 seconds
```



```

tom@snort:~$ ./start_snort.sh
04/03-20:02:04.391482  [**] [1:469:3] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Priority: 2] {ICMP} 192.168.2.1 -> 192.168.1.2
04/03-20:02:04.391482  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.2.1 -> 192.168.1.2
04/03-20:02:04.391603  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1
04/03-20:02:04.392825  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.2.1 -> 192.168.1.2
04/03-20:02:04.392910  [**] [1:451:5] ICMP Timestamp Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1
04/03-20:02:07.380308  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.1:60102 -> 192.168.1.2:161
04/03-20:02:11.129612  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.1:60102 -> 192.168.1.2:705

```

8. Configure Snort to Differentiate Between Internal and External Access

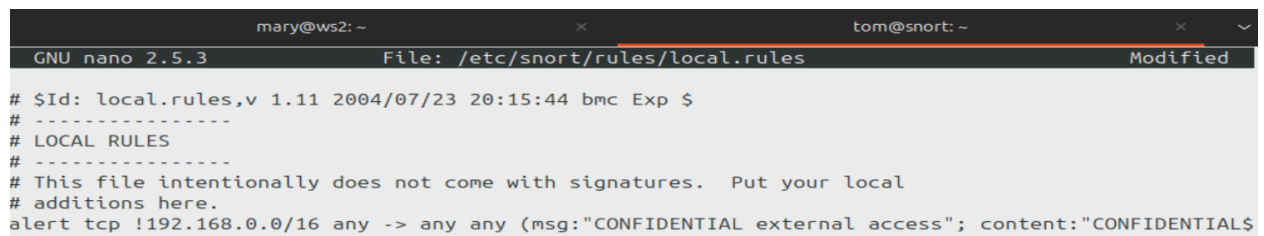
1. Update Snort Rules for Internal and External Access:

Modify the rule in local.rules to trigger alerts only when external addresses access the "CONFIDENTIAL" content:

```

alert tcp !192.168.0.0/16 any -> any any (msg:"CONFIDENTIAL external access";
content:"CONFIDENTIAL"; sid:00004;)

```



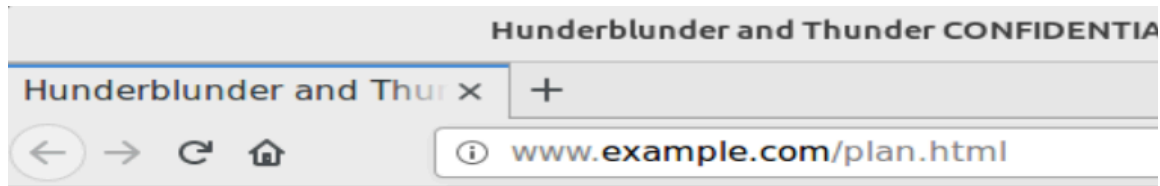
```

mary@ws2: ~                               tom@snort: ~
GNU nano 2.5.3                             File: /etc/snort/rules/local.rules Modified
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert tcp !192.168.0.0/16 any -> any any (msg:"CONFIDENTIAL external access"; content:"CONFIDENTIAL$

```

Restart Snort and test each scenario:

External access to plan.html should trigger an alert.



Hunderblunder and Thunder

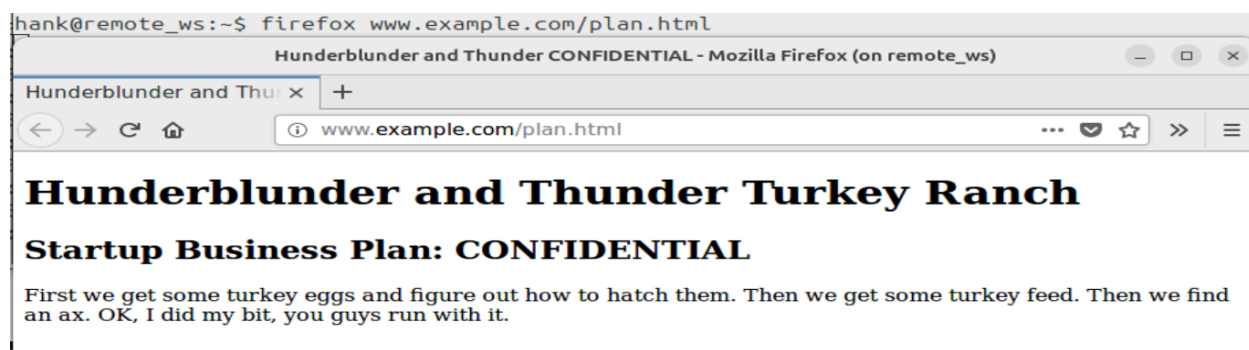
Startup Business Plan: CONFIDENTIAL

First we get some turkey eggs and figure out how to hatch them with an ax. OK, I did my bit, you guys run with it.

```
mary@ws2:~$ firefox www.example.com/plan.html
```

```
04/03-20:11:40.303867  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:41.341235  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:42.107775  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:45.419290  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:45.447513  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:45.561267  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:46.344821  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:47.112621  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:50.421862  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:50.453025  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:50.562962  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:51.345660  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
```

Internal access to plan.html should not.



```
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:35.606734  [**] [1:402:7] ICMP Destination Unreachable [**] [Classifica
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:37.101370  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifica
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:42.107889  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifica
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
```

Both internal and external nmap scans should generate alerts.

```
hank@remote_ws:~$ sudo nmap www.example.com

Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-03 20:15 UTC
Nmap scan report for www.example.com (203.0.113.10)
Host is up (0.00075s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 9.51 seconds
```

```
tom@snort: ~
mary@ws2: ~
tom@snort: ~

04/03-20:12:25.598882  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:27.090963  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:30.602653  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:32.096579  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:35.606734  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:37.101370  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:42.107889  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classification: M
isc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:15:42.361959  [**] [1:469:3] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Pri
ority: 2] {ICMP} 203.0.113.20 -> 203.0.113.10
04/03-20:15:42.361959  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 2
03.0.113.20 -> 203.0.113.10
04/03-20:15:42.363473  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority
: 3] {ICMP} 203.0.113.20 -> 203.0.113.10
04/03-20:15:43.698822  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak]
[Priority: 2] {TCP} 203.0.113.20:64359 -> 203.0.113.10:161
04/03-20:15:45.153375  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information
Leak] [Priority: 2] {TCP} 203.0.113.20:64359 -> 203.0.113.10:705
```

```

mary@ws2:~$ sudo nmap www.example.com

Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-03 20:17 UTC
Nmap scan report for www.example.com (192.168.1.2)
Host is up (0.00047s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 45.74 seconds

```

```

04/03-20:15:42.361959  [**] [1:409:5] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Priority: 2] {ICMP} 203.0.113.20 -> 203.0.113.10
04/03-20:15:42.361959  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.20 -> 203.0.113.10
04/03-20:15:42.363473  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.20 -> 203.0.113.10
04/03-20:15:43.698822  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 203.0.113.20:64359 -> 203.0.113.10:161
04/03-20:15:45.153375  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 203.0.113.20:64359 -> 203.0.113.10:705
04/03-20:17:28.718174  [**] [1:469:3] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Priority: 2] {ICMP} 192.168.2.1 -> 192.168.1.2
04/03-20:17:28.718174  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.2.1 -> 192.168.1.2
04/03-20:17:28.718693  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1
04/03-20:17:28.719754  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.2.1 -> 192.168.1.2
04/03-20:17:28.719837  [**] [1:451:5] ICMP Timestamp Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1
04/03-20:17:30.079169  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.1:48639 -> 192.168.1.2:161
04/03-20:17:30.169900  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.1:48639 -> 192.168.1.2:705

```

9. End the Lab Session

1. Stop the Lab:

In the terminal where you started the lab, type:

```
stoplab snort
```